

Trader Performance Analysis Using Bitcoin Market Sentiment

Assignment Report

Submitted By:
Maryha Sulbhi

Objective

To analyze the relationship between Bitcoin market sentiment and trader performance using Hyperliquid historical trading data and the Bitcoin Fear & Greed Index.

Introduction

Cryptocurrency markets are highly volatile, and trader behavior is often influenced by market sentiment. The Bitcoin Fear & Greed Index measures investor emotions by classifying the market into five categories: **Extreme Fear, Fear, Neutral, Greed, and Extreme Greed**.

This project combines the Bitcoin Fear & Greed Index with Hyperliquid historical trading data to analyze how market sentiment affects trader performance, trading activity, profitability, position size, and buying/selling behavior. The insights obtained from this analysis can help traders make more informed trading decisions and improve risk management strategies.

Objective

The main objectives of this analysis are:

- Analyze trader performance under different market sentiments.
 - Compare profitability across sentiment categories.
 - Examine trading activity during Fear and Greed markets.
 - Study buying and selling behavior.
 - Generate business insights that support smarter trading strategies.
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Dataset Description

Dataset 1: Historical Trader Data

The Hyperliquid historical dataset contains detailed information about executed trades.

Important Features

Feature	Description
Coin	Cryptocurrency traded
Side	Buy/Sell
Size USD	Trade value
Closed PnL	Profit/Loss
Fee	Trading Fee
Timestamp	Trade Time

Dataset 2: Bitcoin Fear & Greed Index

The Bitcoin Fear & Greed dataset provides daily market sentiment.

Possible sentiment classes:

- Extreme Fear
 - Fear
 - Neutral
 - Greed
 - Extreme Greed
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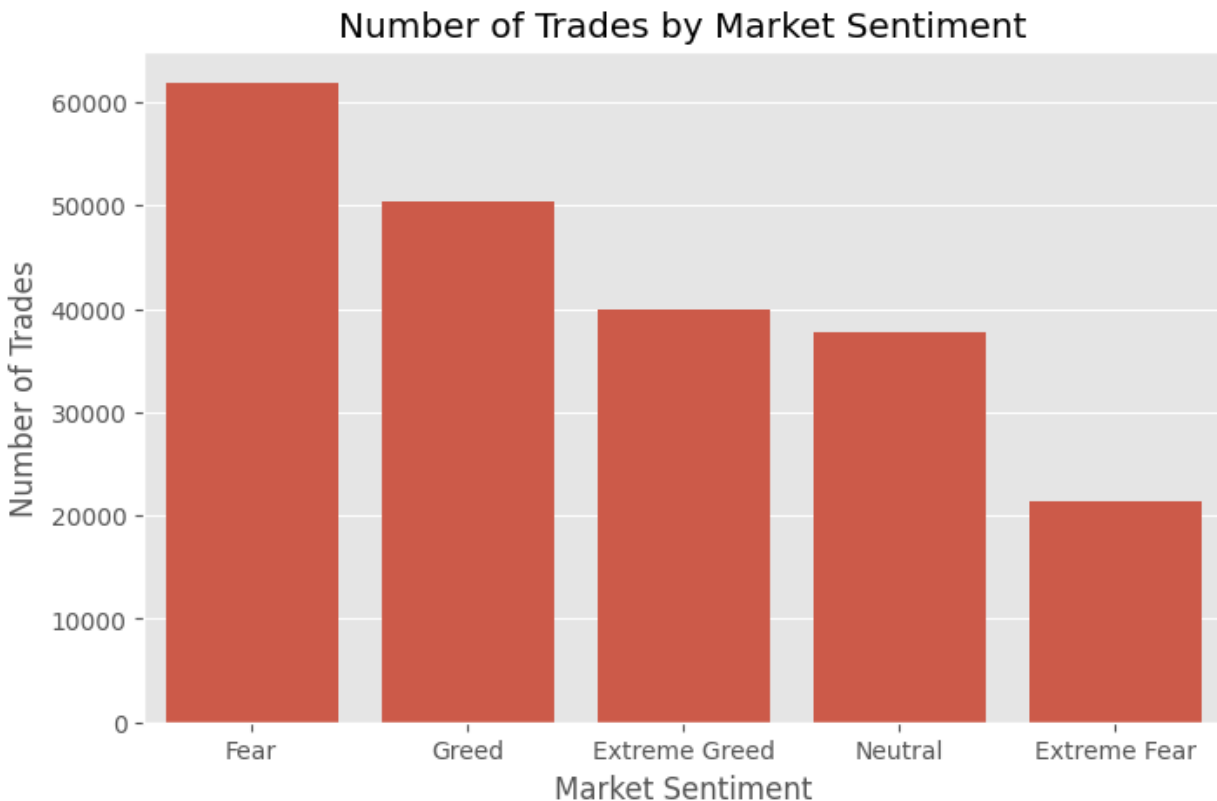
Data Preprocessing

The following preprocessing steps were performed:

- Imported both datasets.
 - Converted timestamps to datetime format.
 - Extracted trading dates.
 - Merged datasets using the Date column.
 - Removed missing values.
 - Created a Trade Result column.
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Exploratory Data Analysis (EDA)

Number of Trades by Market Sentiment



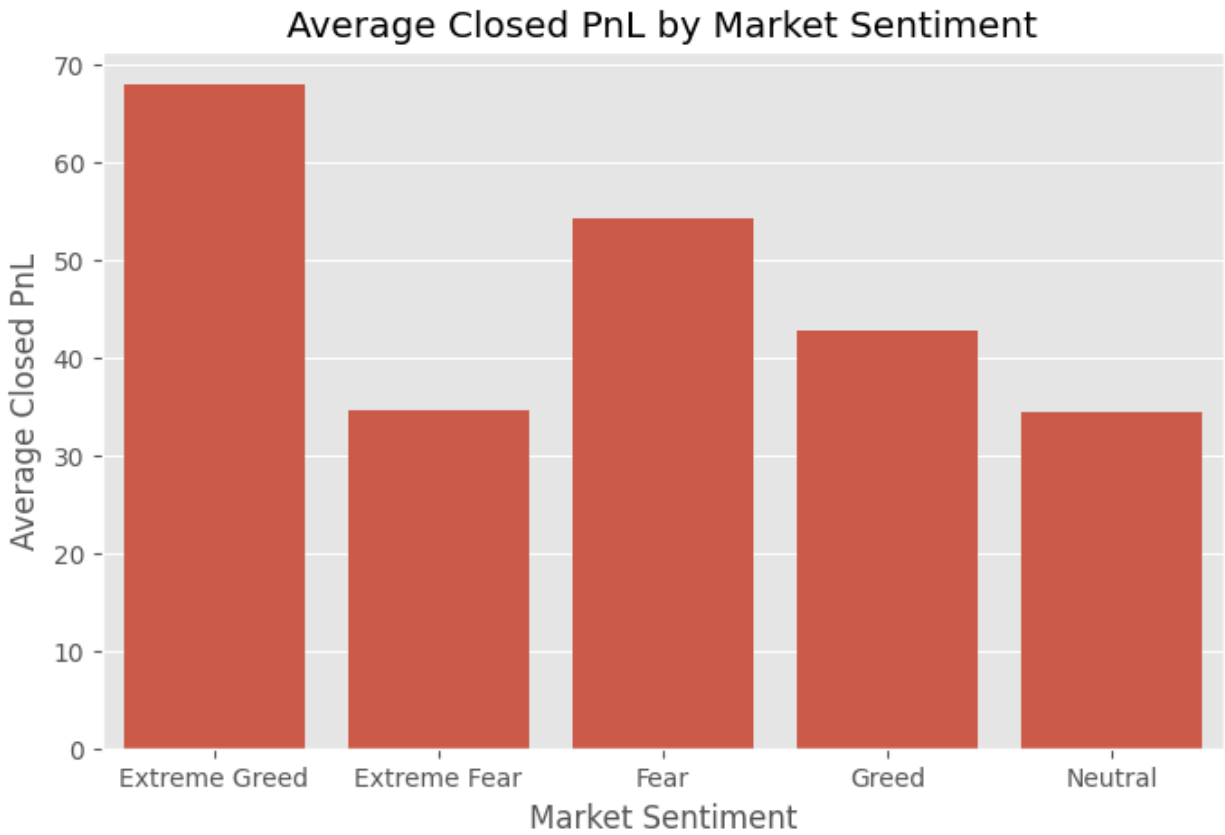
Observation

Fear recorded the highest number of trades, while Extreme Fear had the lowest trading activity.

Insight

Higher market uncertainty encourages increased trading activity, indicating that traders become more active during Fear periods.

Average Closed PnL by Market Sentiment



Observation

Extreme Greed recorded the highest average Closed PnL, while Extreme Fear and Neutral had the lowest.

Insight

Trader profitability tends to improve during Extreme Greed, suggesting that positive market sentiment is associated with better average returns.

Closed PnL Distribution



Observation

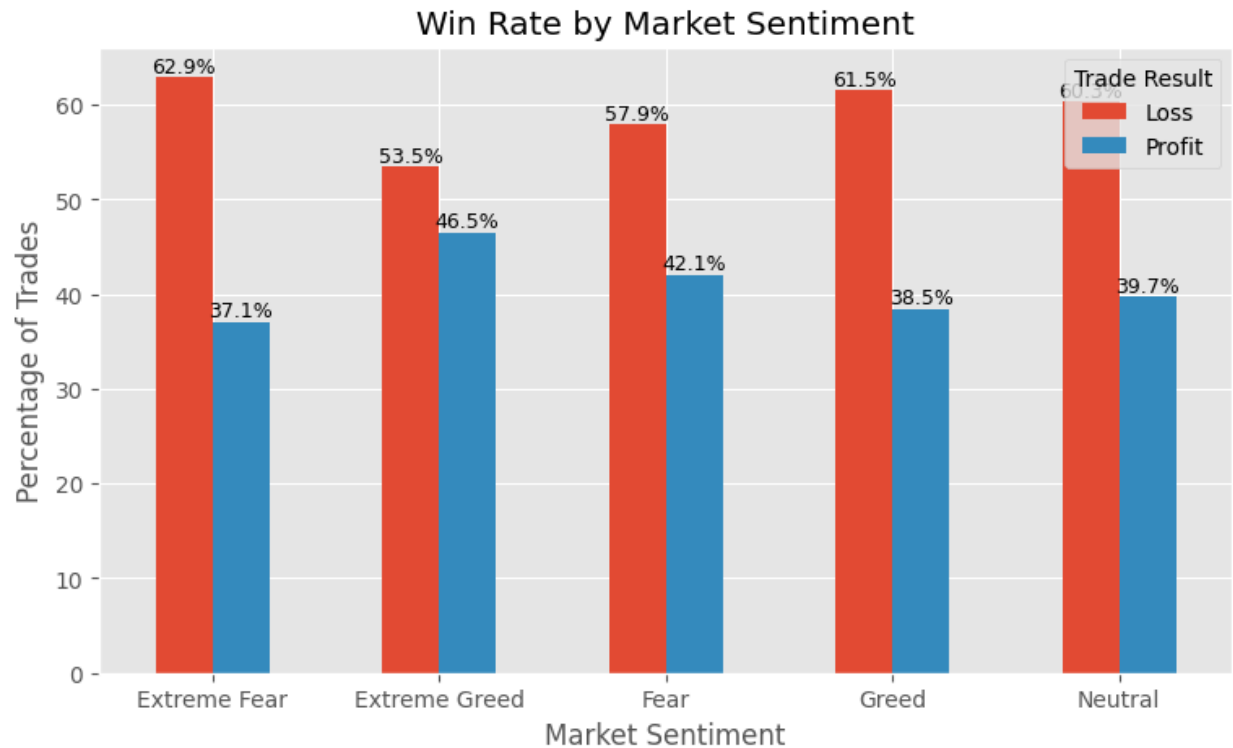
Most trades are concentrated around zero profit or loss, while all market sentiments contain several extreme outliers.

Insight

Although most trades generate small profits or losses, cryptocurrency markets remain highly volatile, with occasional very large gains and losses.

Market Sentiment Analysis

Win Rate by Market Sentiment



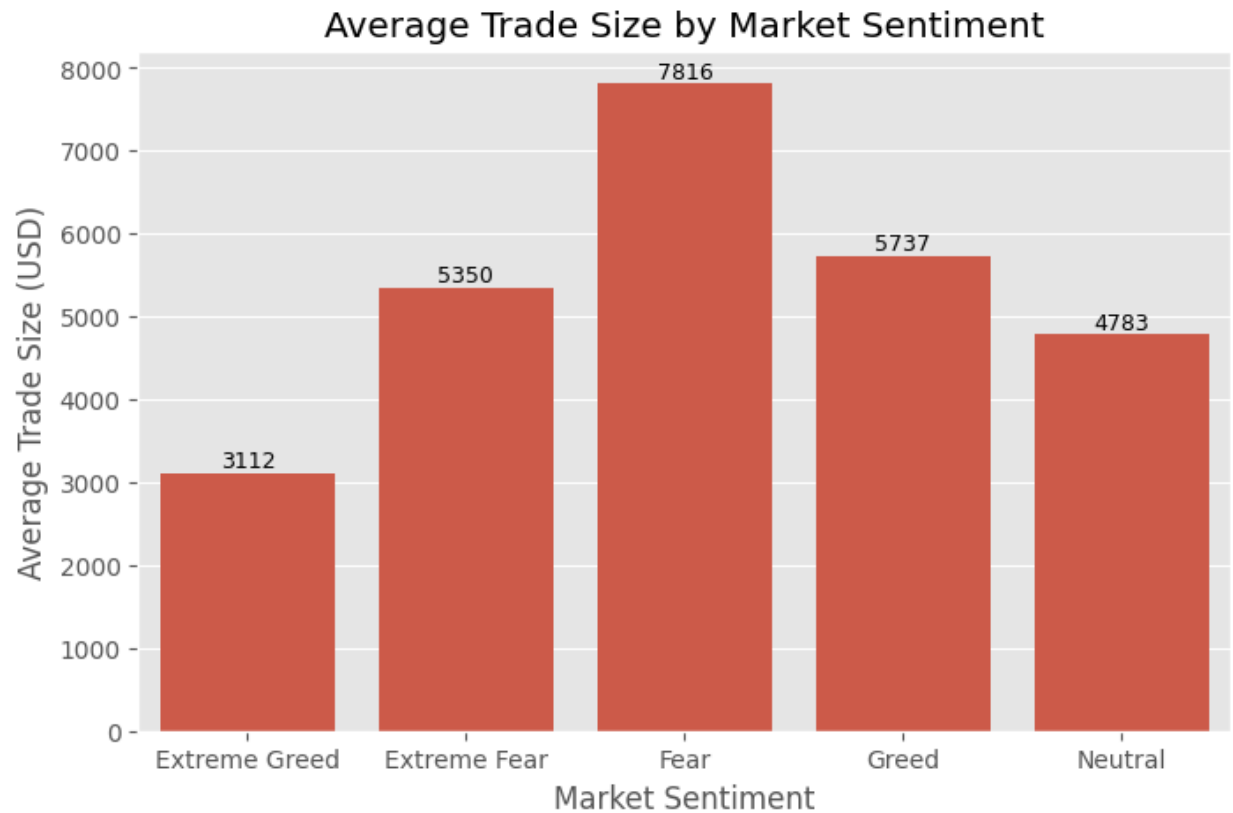
Observation

Extreme Greed achieved the highest win rate (46.5%), while Extreme Fear had the lowest (37.1%).

Insight

Positive market sentiment appears to improve trading success, whereas fearful market conditions are associated with a higher proportion of losing trades.

Average Trade Size



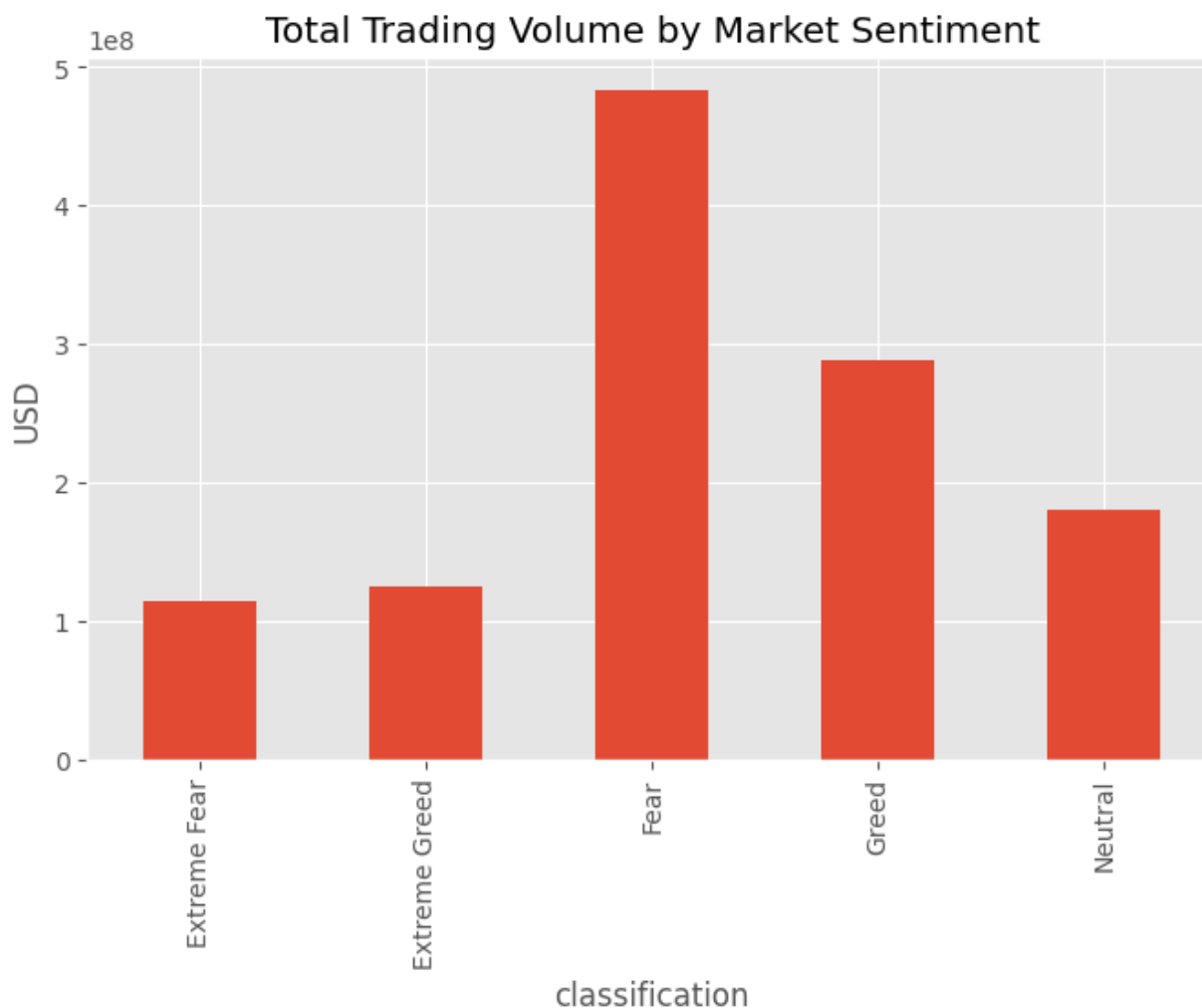
Observation

Fear recorded the highest average trade size, while Extreme Greed had the lowest.

Insight

Traders tend to take larger positions during Fear periods, likely due to increased market volatility and opportunities.

Buy vs Sell Across Market Sentiment



Observation

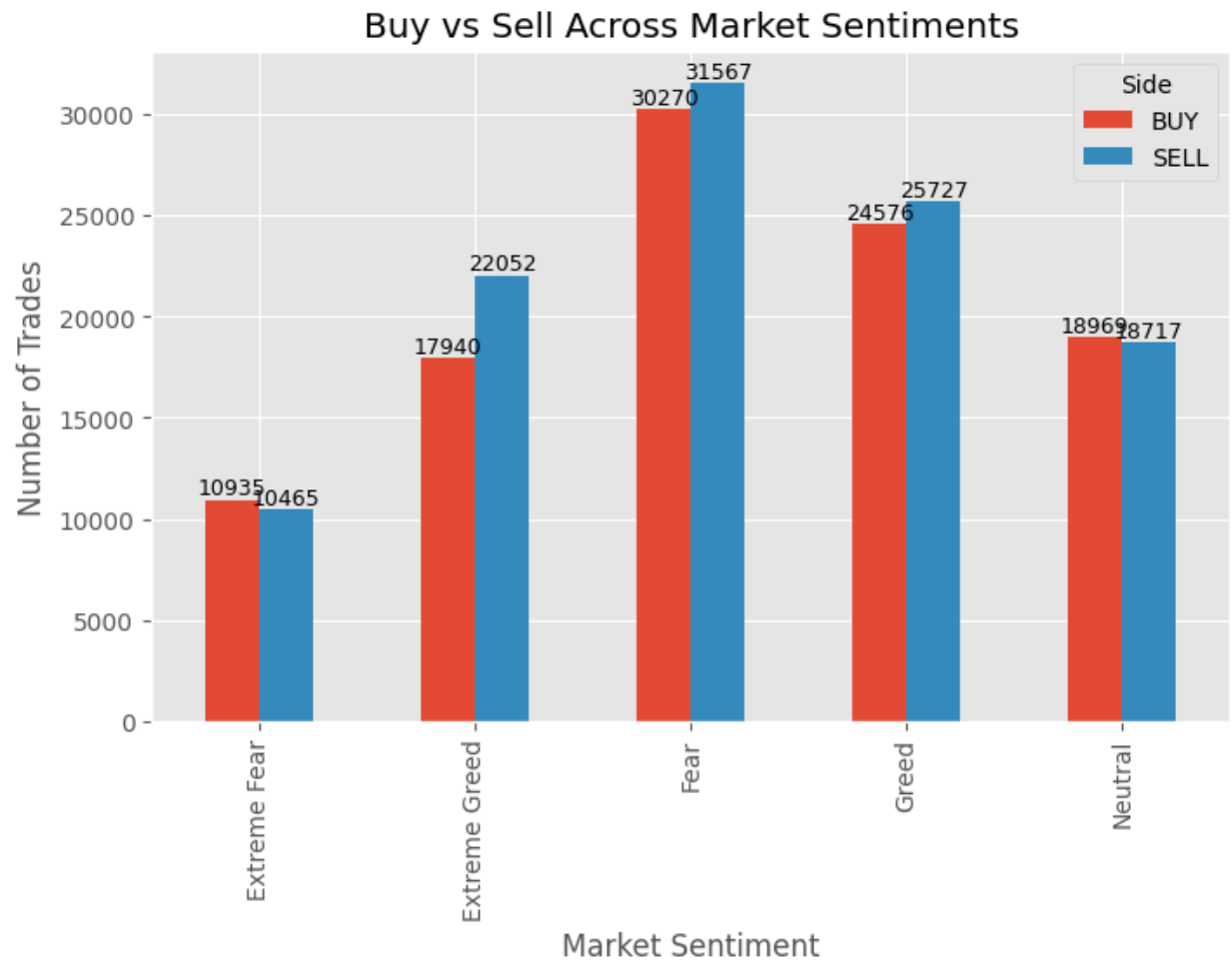
Fear recorded the highest BUY and SELL activity, while Neutral markets showed nearly equal BUY and SELL trades.

Insight

Market sentiment influences trading behavior, with Fear leading to greater participation and Neutral markets showing balanced trading activity.

Additional Analysis

Overall Buy vs Sell Distribution



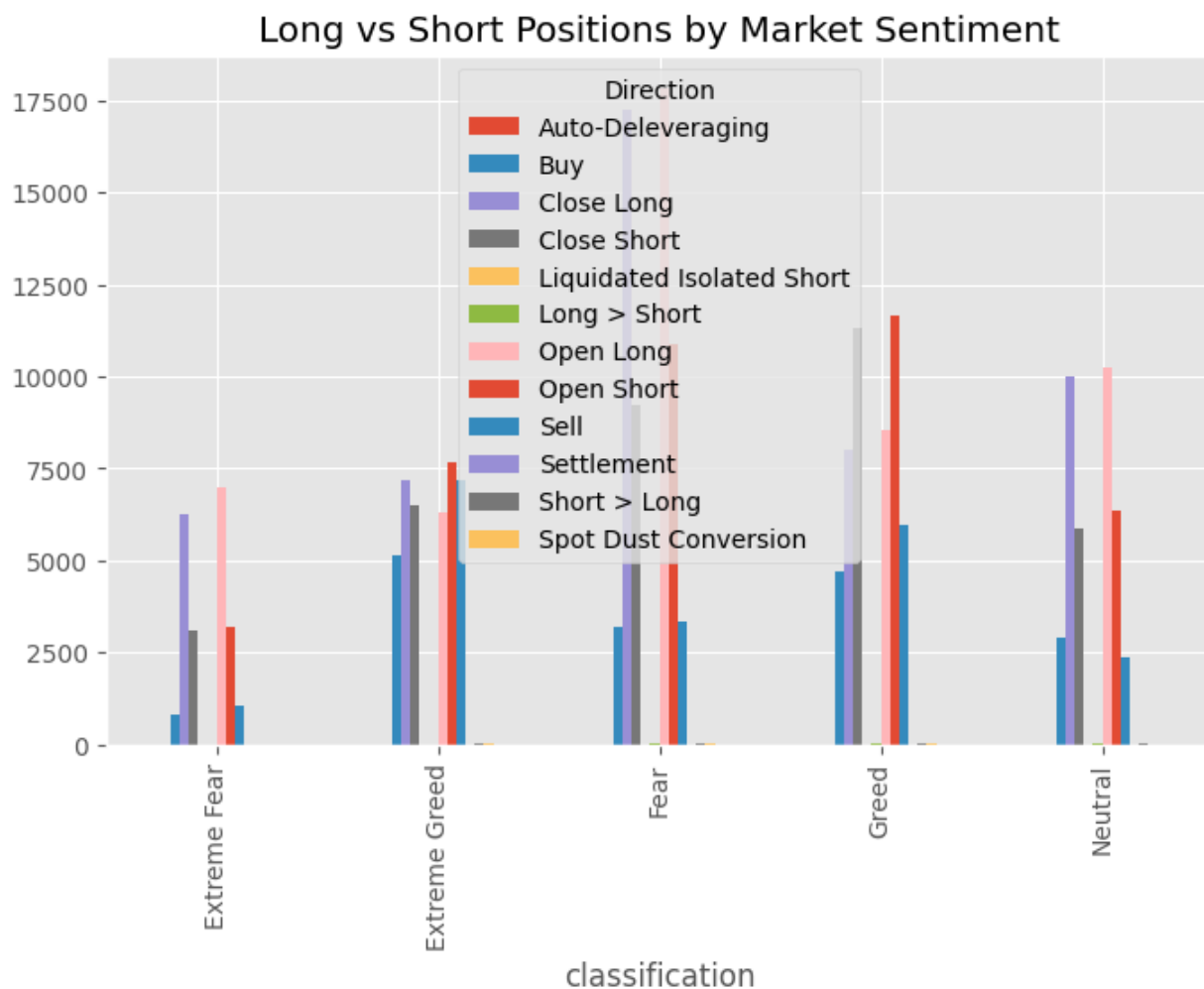
Observation

SELL trades slightly outnumber BUY trades.

Insight

The market exhibits balanced participation from buyers and sellers, with a slight preference for selling.

Trading Fee Distribution



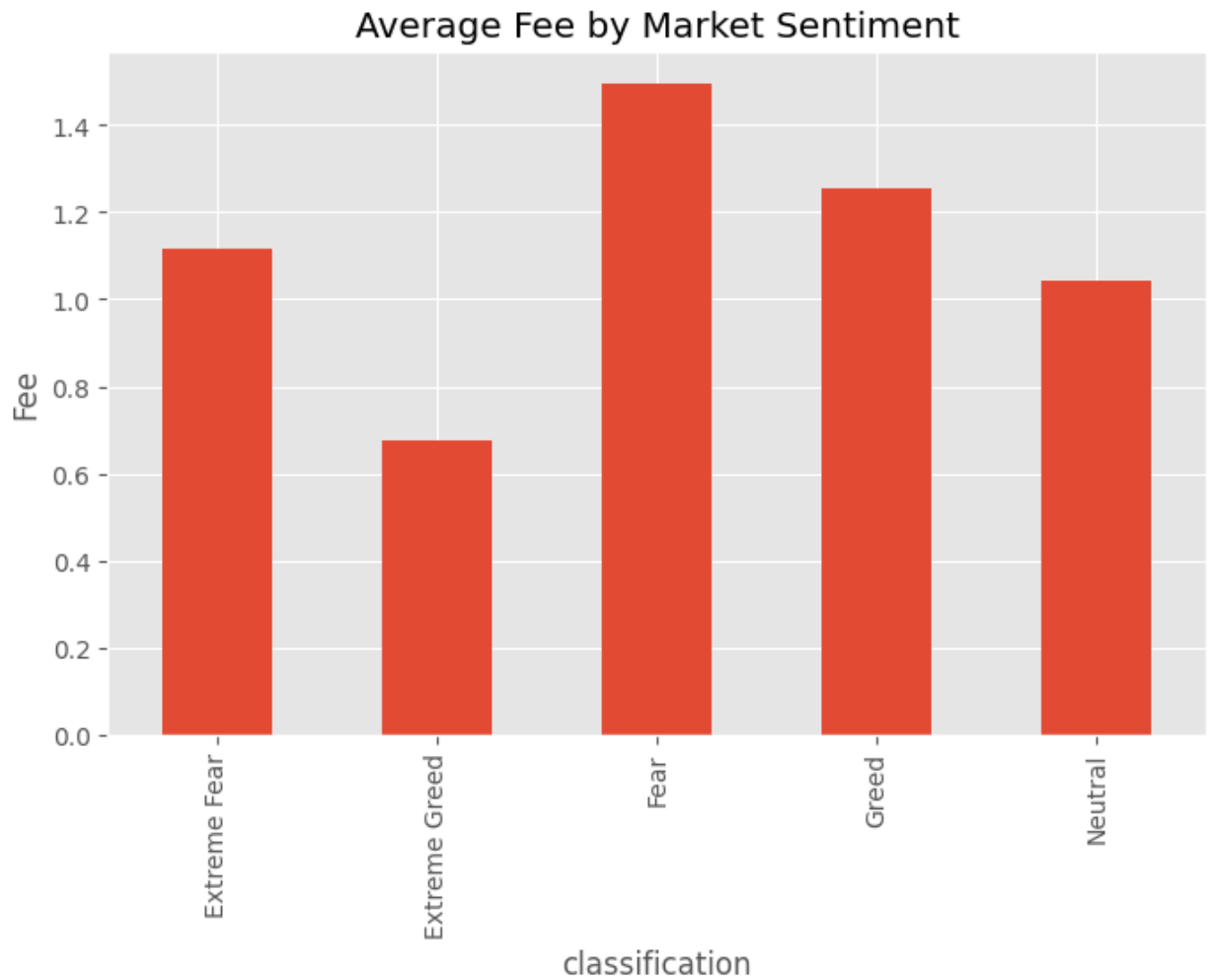
Observation

Most trades incur very low fees, while only a small number of trades have high transaction costs.

Insight

Trading fees remain low for the majority of traders, with high fees primarily associated with large-value trades.

Top Traded Coins

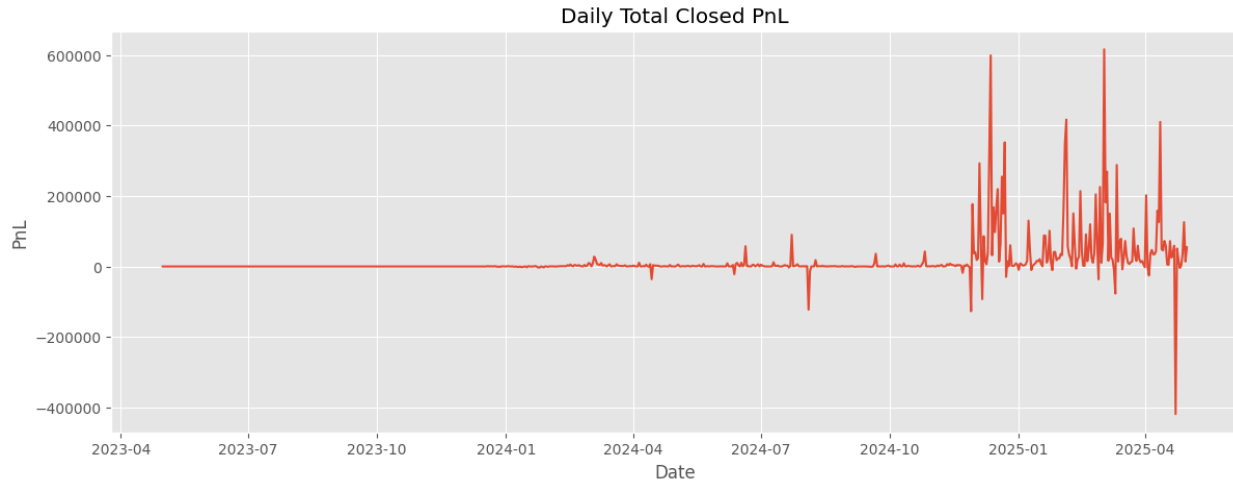


Observation

Trading activity is concentrated among a small number of cryptocurrencies.

Insight

Highly traded assets attract greater liquidity and market participation.



Overall Findings

Key Findings

The analysis revealed the following:

- Fear generated the highest trading activity.
- Extreme Greed achieved the highest average trader profitability.
- Extreme Greed also recorded the highest win rate.
- Traders executed larger trades during Fear periods.
- SELL trades slightly outnumbered BUY trades.
- Most trades incurred very low transaction fees.
- Trading activity was concentrated in a few highly liquid cryptocurrencies.

Business Recommendations

Based on the findings:

- Use market sentiment as an additional trading indicator.
- Apply stricter risk management during Fear periods due to increased volatility.
- Focus on highly liquid cryptocurrencies for better execution.
- Combine sentiment analysis with technical indicators rather than relying solely on market emotions.
- Monitor trading fees, especially for high-value transactions.

Conclusion

This analysis demonstrates that Bitcoin market sentiment has a measurable impact on trader behavior and performance. Fear periods are associated with higher trading activity and larger trade sizes, while Extreme Greed corresponds to higher average profitability and improved win rates.

Although market sentiment alone cannot guarantee profitable trading, it provides valuable information that can support better decision-making when combined with technical analysis and effective risk management.